

A Contractively Complemented Approximate Identity: Literature Resolution of arXiv:math/0512417

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Status. `literature_already_answered`. This note records an explicit later resolution; it does not claim a new theorem.

Source question

Blecher, Hay, and Neal ask in Section 6 of [1] (arXiv PDF page 16; published page 351):

Does every approximately unital operator algebra A have an approximate identity of the form $(1 - x_t)$ with $x_t \in \text{Ball}(A^1)$?

Here A^1 is the unitization. They explain that this is equivalent to the question whether every approximate p -projection is a p -projection.

Decisive later answer

Blecher and Read explicitly state that their Theorem 1.1 was “prompted by, and solves” the question on page 351 of [1]. Their theorem says that every operator algebra with a contractive approximate identity has one, say (e_t) , satisfying

$$\|1 - e_t\| \leq 1, \quad \text{and even} \quad \|1 - 2e_t\| \leq 1 \quad \text{for every } t[2, \textit{Theorem 1.1}].$$

Set $x_t = 1 - e_t \in A^1$. Then $e_t = 1 - x_t$ and $\|x_t\| = \|1 - e_t\| \leq 1$, exactly as requested. The stronger second inequality also implies the first bound by

$$x_t = \frac{1 + (1 - 2e_t)}{2}.$$

The same paper states the equivalent geometric conclusion: every closed projection in A^{**} is a p -projection and is a strong limit of a decreasing net of peak projections [2, Theorem 1.2].

Provenance and scope

This is an exact, author-recognized answer rather than an implication first noticed here: the later paper cites the original source and identifies its page number. It settles both the displayed approximate-identity question and the equivalent p -projection question discussed with it. No residual part of that extracted question remains open. Other broad questions in the source paper, such as general proximality of right ideals, are separate and are not claimed to be settled by this identification.

References

- [1] D. P. Blecher, D. M. Hay, and M. Neal, *Hereditary subalgebras of operator algebras*, J. Operator Theory 59 (2008), 333–357; arXiv:math/0512417.
- [2] D. P. Blecher and C. J. Read, *Operator algebras with contractive approximate identities*, arXiv:1011.3797 (2010, revised 2011).